

Estimation of Copula Models with Discrete Margins via Bayesian Data Augmentation

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Abstract

Estimation of copula models with discrete margins can be difficult beyond the bivariate case. We show how this can be achieved by augmenting the likelihood with latent variables, and computing inference using the resulting augmented posterior. To evaluate this we propose two efficient Markov chain Monte Carlo sampling schemes. One generates the latent variables as a block using a Metropolis-Hasting step with a proposal that is close to its target distribution, the other generates them one at a time. Our method applies to all parametric copulas where the conditional copula functions can be evaluated, not just elliptical copulas as in much previous work. Moreover, the copula parameters can be estimated joint with any marginal parameters, and Bayesian selection ideas employed. We establish the effectiveness of the estimation method by modeling consumer behavior in online retail using Archimedean and Gaussian copulas. The example shows that elliptical copulas can be poor at modeling dependence in discrete data, just as they can be in the continuous case. To demonstrate the potential in higher dimensions we estimate 16 dimensional D-vine copulas for a longitudinal model of usage of a bicycle path in the city of Melbourne, Australia. The estimates reveal an interesting serial dependence structure that can be represented in a parsimonious fashion using Bayesian selection of independence pair-copula components. Finally, we extend our results and method to the case where some margins are discrete and others continuous. Supplemental materials for the article are also available online.

Key Words: Archimedean Copula; Bayesian Pair-Copula Selection; Discrete Longitudinal Data; Markov chain Monte Carlo; Multivariate Discrete Data; Multivariate Dependence; Vine Copulas.

1 Introduction

Copulas have proven a very successful way of modeling dependence in multivariate models. They are now used in a diverse range of applications, proving particularly popular in survival analysis (Clayton 1978; Oakes 1989), finance (Cherubini et al. 2004; McNeil, Frey & Embrechts 2005; Patton 2006) and actuarial science (Frees & Valdez 1998). In the vast majority of instances, parametric copula functions are employed in models for continuous data. In this case the copula parameters, and any marginal model parameters, can often be estimated using maximum likelihood or other methods. However, estimation is more difficult in the case when the data are discrete. Genest & Nešlehová (2007) show that rank-based estimators can be highly erroneous and should not be used, while maximum likelihood estimation can present computational problems. This has limited the use of copulas in fields where multivariate discrete data are common, such as marketing (Danaher & Smith 2011), economics (Cameron et al. 2004) and transport studies (Smith & Kauermann 2011). We address this problem here by presenting an efficient method to compute likelihood-based inference for parametric copula models with discrete margins, or when there is a mixture of discrete and continuous margins.

We specify a joint distribution for the discrete random vector augmented with a vector of latent variables. The latent variable vector has the copula as its marginal distribution, but has a multivariate truncated distribution conditional upon the discrete vector. We show that the resulting margin in the discrete random vector has the probability mass function of the copula model. Finite sample inference on the copula parameters, and any marginal model parameters, is obtained from the resulting augmented posterior distribution. This is evaluated efficiently using one of two alternative Markov chain Monte Carlo (MCMC) sampling schemes. The first is where the elements of the latent vector are generated one at a time, and is an extension of the sampler suggested by Pitt, Chan & Kohn (2006) to non-elliptical copulas. The other generates the latent vector as a block using a Metropolis-Hastings step with a proposal that is close to the target distribution, but from which it is faster to generate. The method can be used in high dimensions, and for any copula where conditional copula distribution functions can be evaluated, which includes all popular

parametric copulas employed currently. The finite sample distribution of different measures of dependence for the fitted copula model can be computed from the Monte Carlo sample. We also show how to extend both the results, and the two sampling schemes, to the case where some margins are discrete and others continuous.

We demonstrate our approach using two examples. The first is a bivariate marketing study of online consumer behavior at amazon.com. We show that the level of exposure to the website during a visit is positively related to both the amount spent and purchase incidence. In both cases the dependence is asymmetric and captured better by one and two parameter Archimedean copulas than a Gaussian copula with symmetric dependence. Elliptical copulas are known to be inadequate models of dependence for much continuous data (Patton 2006), and the example demonstrates this can also be true for discrete data. We also show here that ignoring the discrete nature of the data, and treating it as continuous, gives erroneous estimates in a manner similar to rank-based estimators as identified by Genest & Nešlehová (2007).

A new and flexible copula for higher dimensional data is the D-vine (Joe 1996; Bedford & Cooke 2002; Aas et al. 2009; Min & Czado 2010; Haff, Aas & Frigessi 2010), which is constructed from a sequence of bivariate ‘pair-copulas’. Our second example is a 16 dimensional longitudinal D-vine copula model for the number of bicycles travelling down a bike path in the city of Melbourne. Each margin corresponds to the count of the number of bicycles that pass each hour. Smith et al. (2010) show that a D-vine copula model is well-motivated for the analysis of serial dependence. They also show how to use Bayesian selection to identify independence pair-copula components of the vine, and we extend their method to the discrete case here. The bike path is mainly used by commuters, and an interesting sparse dependence structure is uncovered. We evaluate the bivariate margin in the morning and afternoon peak hours from fitted D-vines with Gumbel and t pair-copulas, and find that there is positive nonlinear dependence. We show that the choice of copula is important by comparing forecasts of the afternoon peak counts, given the morning peaks.

Estimating equation methods can be used to estimate multivariate models for discrete data (Liang, Zeger & Qaquis 1992). However, Song, Li & Yuan (2009) list a number

of shortcomings of this approach compared to maximum likelihood estimation, including a loss in efficiency. Nevertheless, direct maximization of the likelihood function can be difficult here, even for a small number of dependence parameters; for example, Song et al. (2009) and Nikoloulopoulos & Karlis (2008) only employ 3 parameters. In comparison, Bayesian data augmentation provides full likelihood-based inference for a much larger number of dependence parameters, with up to 240 in our D-vine example. Also, much previous work has focused specifically on Gaussian copula models (Song et al. 2009; Madsen & Fang 2010; Pitt et al. 2006), whereas our approach applies to other parametric copula functions. We note that Pitt et al. (2006) propose a Bayesian data augmentation method with Gaussian latent variables, which is shown to work in a 45 dimensional example in Danaher & Smith (2011). Our paper extends this approach to non-elliptical copulas, provides the distributional theory for data augmentation, proposes new sampling schemes, and demonstrates the usefulness of the methodology in a number of contemporary applications where the dependence structure is too complex to be captured by elliptical copulas.

The rest of the paper is organized as follows. Section 2 outlines the distribution of the discrete vector, augmented with the latent variable vector. The conditional distributions of the latents are derived, which are used to develop two MCMC sampling schemes in Section 3. It is shown how these can be used to compute posterior inference, including dependence measures from the fitted copula model. Section 4 contains the online retail example, while Section 5 shows how to employ the approach to estimate a D-vine copula, including selection of component pair-copulas. Section 6 extends the methodology to the mixed data case, and Section 7 discusses some advantages of the approach.

2 Copula Models for Discrete Data

2.1 Copula Function

The function $C(u_1, \dots, u_m)$ is called a copula function if it is a distribution function with each of its margins uniformly distributed on $[0, 1]$. That is, $C(\mathbf{u}) = \Pr(U_1 \leq u_1, \dots, U_m \leq u_m)$, with each U_j , for $j = 1, \dots, m$, uniformly distributed on $[0, 1]$ and $\mathbf{u} = (u_1, \dots, u_m)$. The density $c(\mathbf{u}) = \partial C(\mathbf{u}) / (\partial \mathbf{u})$ is called the copula density when it exists. Joe (1997) and

Nelsen (2006) discuss a wide range of choices for C and their properties.

Following Sklar (1959) a joint distribution function F with marginal distribution functions $F_1, \dots, F_m$ can be written as

$$F(\mathbf{x}) = C(F_1(x_1), \dots, F_m(x_m)), \quad (2.1)$$

with $\mathbf{x} = (x_1, \dots, x_m)$. When $F_1, \dots, F_m$ are strictly monotonically increasing, so that the margins are continuous, C is known to be unique. However, when one or more marginal distribution is discrete, this is no longer the case; see Genest & Nešlehová (2007). Nevertheless, the copula model in equation (2.1) remains a well-defined distribution function F for any admissible copula function C . Moreover, in applied modeling C is usually picked from a parametric family, and F defined in this manner; for example, see Cameron et al. (2004).

2.2 Augmented Distribution

Consider the case where $\mathbf{X} = (X_1, \dots, X_m)$ has discrete-valued elements and distribution function F at equation (2.1). Let $b_j = F_j(x_j)$, and $a_j = F_j(x_j^-)$ be the left hand limit of F_j at x_j , so that $a_j = F_j(x_j - 1)$ if X_j is ordinal. Then, the probability mass function of $\mathbf{X}$ can be expressed in closed form as

$$f(\mathbf{x}) = \Pr(X_1 = x_1, \dots, X_m = x_m) = \Delta_{a_1}^{b_1} \Delta_{a_2}^{b_2} \cdots \Delta_{a_m}^{b_m} C(\mathbf{v}), \quad (2.2)$$

where $\mathbf{v} = (v_1, \dots, v_m)$ and we employ the difference notation of Nelsen (2006; p.43):

$$\begin{aligned} \Delta_{a_k}^{b_k} C(u_1, \dots, u_{k-1}, v_k, u_{k+1}, \dots, u_m) = \\ C(u_1, \dots, u_{k-1}, b_k, u_{k+1}, \dots, u_m) - C(u_1, \dots, u_{k-1}, a_k, u_{k+1}, \dots, u_m), \end{aligned}$$

with v_k an index of differencing. For example, when $m = 3$,

$$\begin{aligned} f(x_1, x_2, x_3) &= \Delta_{a_1}^{b_1} \Delta_{a_2}^{b_2} \Delta_{a_3}^{b_3} C(v_1, v_2, v_3) \\ &= C(b_1, b_2, b_3) - C(b_1, b_2, a_3) - C(b_1, a_2, b_3) - C(a_1, b_2, b_3) \\ &\quad + C(b_1, a_2, a_3) + C(a_1, b_2, a_3) + C(a_1, a_2, b_3) - C(a_1, a_2, a_3). \end{aligned}$$

In general, estimation of any copula parameters for C using direct maximum likelihood estimation is difficult for two reasons. First, there are 2^m terms in the sum at equation (2.2), so that to compute the likelihood for n observations involves $n2^m$ evaluations of C , which is prohibitive for larger values of m . Second, even in the case when $m > 2$ is small the likelihood can prove difficult to maximise for some copula models, particularly when the copula and marginal parameters are estimated jointly.

We instead consider the joint distribution of $(\mathbf{X}, \mathbf{U})$, with $\mathbf{U} = (U_1, \dots, U_m)$. To express this, first note that F_j is a many-to-one function and $X_j|U_j$ is a degenerate distribution with density $f(x_j|u_j) = \mathcal{I}(F_j(x_j^-) \leq u_j < F_j(x_j))$. Here, the indicator function $\mathcal{I}(A) = 1$ if A is true, and zero otherwise.¹ Then $(\mathbf{X}, \mathbf{U})$ has mixed probability density

$$f(\mathbf{x}, \mathbf{u}) = f(\mathbf{x}|\mathbf{u})c(\mathbf{u}) = \prod_{j=1}^m \mathcal{I}(F_j(x_j^-) \leq u_j < F_j(x_j))c(\mathbf{u}), \quad (2.3)$$

where $f(\mathbf{x}|\mathbf{u}) = \prod_{j=1}^m f(x_j|u_j)$.

Proposition 1

If $(\mathbf{X}, \mathbf{U})$ has mixed probability density given by equation (2.3), then the marginal probability mass function of $\mathbf{X}$ is given by equation (2.2).

Proof : See Appendix.

2.3 Latent Variable Distributions

We show in Section 3 how to estimate the copula model using the likelihood augmented with latent variables distributed as $\mathbf{U}$. The computations are undertaken using Markov chain Monte Carlo (MCMC) algorithms. To develop these, the conditional distributions of the latent variables require evaluation. From equation (2.3) the density of $\mathbf{U}|\mathbf{X}$ is

$$f(\mathbf{u}|\mathbf{x}) = \frac{c(\mathbf{u})}{f(\mathbf{x})} \prod_{j=1}^m \mathcal{I}(a_j \leq u_j < b_j), \quad (2.4)$$

¹An alternative notation is $f(x_j|u_j) = \delta_{x_j}(F_j^-(u_j))$, which is a Dirac mass at $F_j^-(u_j)$, with F_j^- the quantile function.

so that $\mathbf{U}|\mathbf{X}$ is truncated to $[a_1, b_1) \times \cdots \times [a_m, b_m)$. However, for a subset of elements of $\mathbf{U}$ the conditional distribution is more complex.

Proposition 2

For $j = 1, \dots, m-1$ the density of $(U_1, \dots, U_j)|\mathbf{X}$ is

$$f(u_1, \dots, u_j|\mathbf{x}) = \frac{c(u_1, \dots, u_j)}{f(\mathbf{x})} \left(\Delta_{a_{j+1}}^{b_{j+1}} \cdots \Delta_{a_m}^{b_m} C_{j+1, \dots, m|1, \dots, j}(v_{j+1}, \dots, v_m|u_1, \dots, u_j) \right) \prod_{k=1}^j \mathcal{I}(a_k \leq u_k < b_k),$$

where $c(u_1, \dots, u_j) = \int c(\mathbf{u}) du_{j+1} \cdots du_m$, $C_{j+1, \dots, m|1, \dots, j}$ is the distribution function of $U_{j+1}, \dots, U_m|U_1, \dots, U_j$, and $v_{j+1}, \dots, v_m$ are indices of differencing.

Proof : See Appendix.

Here, $c(u_1, \dots, u_j)$ is the marginal copula density with support on $[0, 1]^j$, while Proposition 2 shows that $f(u_1, \dots, u_j|\mathbf{x})$ has support on $[a_1, b_1) \times \cdots \times [a_j, b_j)$. That is, $(U_1, \dots, U_j)$ is truncated conditional on $\mathbf{X}$. The density has a normalizing constant which involves the summation of 2^{m-j} terms, and when $j = m$, the density at equation (2.4) results. Throughout the paper if $I_1 \subset \{1, \dots, m\}$, $I_2 \subset \{1, \dots, m\}$ and $I_1 \cap I_2 = \emptyset$, then we denote the conditional distribution and density functions of $\{U_j; j \in I_1\}|\{U_k; k \in I_2\}$ as $C_{I_1|I_2}$ and $c_{I_1|I_2}$, respectively. The corollary below is used in developing the MCMC algorithms.

Corollary 1

For $j = 2, \dots, m$ the conditional distribution of $U_j|U_1, \dots, U_{j-1}, \mathbf{X}$ is

$$f(u_j|u_1, \dots, u_{j-1}, \mathbf{x}) = c_{j|1, \dots, j-1}(u_j|u_1, \dots, u_{j-1}) \mathcal{I}(a_j \leq u_j < b_j) \mathcal{K}_j(u_1, \dots, u_j),$$

where $\mathcal{K}_m(\mathbf{u}) = 1 / (\Delta_{a_m}^{b_m} C_{m|1, \dots, m-1}(v_m|u_1, \dots, u_{m-1}))$, and for $j = 2, \dots, m-1$:

$$\mathcal{K}_j(u_1, \dots, u_j) = \frac{\Delta_{a_{j+1}}^{b_{j+1}} \cdots \Delta_{a_m}^{b_m} C_{j+1, \dots, m|1, \dots, j}(v_{j+1}, \dots, v_m|u_1, \dots, u_j)}{\Delta_{a_j}^{b_j} \cdots \Delta_{a_m}^{b_m} C_{j, \dots, m|1, \dots, j-1}(v_j, \dots, v_m|u_1, \dots, u_{j-1})}.$$

Proof : Follows immediately from Proposition 2 by considering $f(u_j|u_1, \dots, u_{j-1}, \mathbf{x}) = f(u_1, \dots, u_j|\mathbf{x}) / f(u_1, \dots, u_{j-1}|\mathbf{x})$.

3 Estimation & Inference

3.1 Augmented Likelihood

In applied analysis, a copula function is usually selected from a parametric family and parametric models are often used for the margins. If θ_j are the parameters of margin j , and ϕ are the copula parameters, we denote the marginal distribution functions as $F_j(x_j; \theta_j)$, copula function as $C(\mathbf{u}; \phi)$ and copula density as $c(\mathbf{u}; \phi)$. Consider an independent sample of n observations, each with distribution function given at equation (2.1). Throughout the rest of this section we denote each observation as $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,m})$, and $\mathbf{x} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$. To estimate $\Theta = \{\theta_1, \dots, \theta_m\}$ and ϕ we introduce latent variables $\mathbf{u}_i = (u_{i,1}, \dots, u_{i,m})$, for $i = 1, \dots, n$, with $(\mathbf{x}_i, \mathbf{u}_i)$ having joint density at equation (2.3). The augmented likelihood is

$$f(\mathbf{u}, \mathbf{x} | \Theta, \phi) = \prod_{i=1}^n f(\mathbf{x}_i, \mathbf{u}_i | \Theta, \phi) = \prod_{i=1}^n c(\mathbf{u}_i; \phi) \prod_{j=1}^m \mathcal{I}(a_{i,j} \leq u_{i,j} < b_{i,j}), \quad (3.1)$$

where $a_{i,j} = F_j(x_{i,j}^-; \theta_j)$, $b_{i,j} = F_j(x_{i,j}; \theta_j)$ and $\mathbf{u} = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$. Throughout the rest of this section it is important to remember that $a_{i,j}$ and $b_{i,j}$ are functions of θ_j and $x_{i,j}$.

In some problems, such as in multivariate financial time series models (Patton 2006; Jondeau & Rockinger 2006), the marginal distributions vary over observations. In this case the marginal distribution functions are denoted as $F_{i,j}$, with $a_{i,j} = F_{i,j}(x_{i,j}^-; \theta_j)$ and $b_{i,j} = F_{i,j}(x_{i,j}; \theta_j)$. In other problems the empirical distribution function is employed for the margins (Oakes 1994; Genest et al. 1995)

We assume the prior density $\pi(\Theta, \phi) = \pi(\phi) \prod_{j=1}^m \pi(\theta_j)$. The prior $\pi(\phi)$ is specific to the choice of copula, and $\pi(\theta_j)$ is specific to any marginal model, and both can be chosen by the user.

3.2 Conditional Posterior of Copula Parameters

In both sampling schemes, we generate the copula parameters ϕ conditional upon $\mathbf{u}$ from

$$f(\phi | \mathbf{u}, \Theta, \mathbf{x}) = f(\phi | \mathbf{u}) \propto \prod_{i=1}^n c(\mathbf{u}_i; \phi) \pi(\phi),$$

which greatly simplifies the problem. The manner of generation depends upon the type of copula C , and the prior $\pi(\phi)$. For an elliptical copula this involves generation of a correlation matrix. Pitt et al. (2006) show how to do this for a covariance selection prior, while Danaher & Smith (2011) show how to do it with a prior on a Cholesky factor based decomposition. Other priors for correlation matrices, such as the shrinkage prior in Daniels & Pourahmadi (2009), can also be employed here. For many other copulas and priors ϕ can be generated one element at a time using a Metropolis-Hastings (MH) step with a random walk (Robert & Cassella 2004, pp.287-291) or other proposal. In our empirical work we show that this works well for several Archimedean and D-vine copulas.

3.3 Conditional Posterior of Latent Variables

The posterior $f(\mathbf{u}|\phi, \Theta, \mathbf{x}) = \prod_{i=1}^n f(\mathbf{u}_i|\phi, \Theta, \mathbf{x}_i)$, where $f(\mathbf{u}_i|\phi, \Theta, \mathbf{x}_i)$ has a multivariate truncated density of the form at equation (2.4). This is hard to generate from directly, so that in our first sampling scheme we use a MH algorithm with proposal density $g(\mathbf{u}_i) = \prod_{j=2}^m g_j(u_{i,j}|u_{i,1}, \dots, u_{i,j-1})g_1(u_{i,1})$. Each density g_j is proportional to $c_{j|1,\dots,j-1}$, and truncated to $[a_{i,j}, b_{i,j})$, so that for $j > 1$

$$g_j(u_{i,j}|u_{i,1}, \dots, u_{i,j-1}) = \frac{c_{j|1,\dots,j-1}(u_{i,j}|u_{i,1}, \dots, u_{i,j-1}; \phi) \mathcal{I}(a_{i,j} \leq u_{i,j} < b_{i,j})}{C_{j|1,\dots,j-1}(b_{i,j}|u_{i,1}, \dots, u_{i,j-1}; \phi) - C_{j|1,\dots,j-1}(a_{i,j}|u_{i,1}, \dots, u_{i,j-1}; \phi)}, \quad (3.2)$$

and $g_1(u_{i,1}) = \mathcal{I}(a_{i,1} \leq u_{i,1} < b_{i,1})/(b_{i,1} - a_{i,1})$.

Notice that when $j = m$, from Corollary 1, $g_m(u_{i,m}|u_{i,1}, \dots, u_{i,m-1}) = f(u_{i,m}|u_{i,1}, \dots, u_{i,m-1}, \mathbf{x}_i, \phi, \Theta)$ exactly. For $j < m$, $g_j(u_{i,j}|u_{i,1}, \dots, u_{i,j-1})$ is close to $f(u_{i,j}|u_{i,1}, \dots, u_{i,j-1}, \mathbf{x}_i, \phi, \Theta)$, with the difference being determined by the normalizing constant of g_j and the term $\mathcal{K}_j(u_{i,1}, \dots, u_{i,j})$ defined in Corollary 1. However, as long as $C_{j|1,\dots,j-1}$ and its inverse are fast to compute, generation from g_j is fast; whereas, it is difficult to generate from $f(u_{i,j}|u_{i,1}, \dots, u_{i,j-1}, \mathbf{x}_i, \phi, \Theta)$ directly.

The MH method sequentially generates $u_{i,j}^{\text{new}}$ from g_j , then the vector $\mathbf{u}_i^{\text{new}} = (u_{i,1}^{\text{new}}, \dots, u_{i,m}^{\text{new}})$

is accepted over the previous values $\mathbf{u}_i^{\text{old}} = (u_{i,1}^{\text{old}}, \dots, u_{i,m}^{\text{old}})$ with probability $\min(1, \alpha_i)$, where

$$\alpha_i = \prod_{j=2}^m \frac{C_{j|1,\dots,j-1}(b_{i,j}|u_{i,1}^{\text{new}}, \dots, u_{i,j-1}^{\text{new}}; \phi) - C_{j|1,\dots,j-1}(a_{i,j}|u_{i,1}^{\text{new}}, \dots, u_{i,j-1}^{\text{new}}; \phi)}{C_{j|1,\dots,j-1}(b_{i,j}|u_{i,1}^{\text{old}}, \dots, u_{i,j-1}^{\text{old}}; \phi) - C_{j|1,\dots,j-1}(a_{i,j}|u_{i,1}^{\text{old}}, \dots, u_{i,j-1}^{\text{old}}; \phi)}$$

is derived using Corollary 1. Note that as $(F_j(x; \theta_j) - F_j(x^-; \theta_j)) \rightarrow 0$ for all j (ie. the marginal distributions get closer to being continuous) then $u_{i,j}^{\text{new}} \rightarrow u_{i,j}^{\text{old}}$, so that $\alpha_i \rightarrow 1$. In this sense, $g(\mathbf{u}_i)$ is close to $f(\mathbf{u}_i|\phi, \Theta, \mathbf{x}_i)$, and we show in our empirical work that, even when modeling binary data, the step provides adequate acceptance rates.

We note that to generate from g involves $3(m-1)$ evaluations of the conditional copula distribution functions or their inverses. An additional $2(m-1)$ evaluations are required to compute the denominator of α_i . Therefore, the computational burden of the MH step is less when evaluation of $C_{j|1,\dots,j}$ involves few calculations. Last, it is also possible to generate each $u_{i,j}$ from $f(u_{i,j}|u_{i,1}, \dots, u_{i,j-1}, \mathbf{x}_i, \phi, \Theta)$ separately using g_j as a proposal. However, $\mathcal{K}_j$ needs computing, so that to generate all elements of $\mathbf{u}_i$ involves $O(2^{m-1})$ evaluations of the conditional distribution functions, and is therefore unattractive.

3.4 Sampling Schemes

We propose two MCMC sampling schemes to estimate ϕ and Θ jointly. The first scheme is:

Sampling Scheme 1 (*Blocked latents and marginal parameters*)

- (1) Generate from $f(\Theta|\phi, \mathbf{x})$
- (2) Generate from $f(\mathbf{u}|\Theta, \phi, \mathbf{x})$
- (3) Generate from $f(\phi|\mathbf{u})$

Steps (1) and (2) together are equivalent to generating from $f(\Theta, \mathbf{u}|\phi, \mathbf{x})$ as a block, so that Scheme 1 is likely to exhibit strong convergence and mixing. However, in Step (1)

$$f(\Theta|\phi, \mathbf{x}) \propto f(\mathbf{x}|\Theta, \phi) \prod_{j=1}^m \pi(\theta_j) = \prod_{i=1}^n \Delta_{a_{i,1}}^{b_{i,1}} \dots \Delta_{a_{i,m}}^{b_{i,m}} C(\mathbf{v}; \phi) \prod_{j=1}^m \pi(\theta_j), \quad (3.3)$$

which requires computation of the likelihood $f(\mathbf{x}|\Theta, \phi)$, and is an $O(2^m)$ operation as noted previously. We generate Θ using MH with proposal density $q(\Theta) = \prod_{j=1}^m q_j(\theta_j)$. We fol-

low Chib & Greenberg (1998) and use a multivariate t density for q_j with $\nu = 7$ degrees of freedom. The proposal is centred around the estimate of θ_j obtained by two or three Newton-Raphson steps starting from the marginal model estimate, and with scale equal to the negative of the inverse of the information matrix. We note that in problems where Θ has a large number of elements, it might prove attractive to partition Θ and generate from the resulting margins of $f(\Theta|\phi, \mathbf{x})$. For Step (2) we generate $\mathbf{u}_i$ using the MH step outlined in Section 3.3.

Denote $\mathbf{u}_{(j)} = \{u_{1,j}, \dots, u_{n,j}\}$ and $\mathbf{x}_{(j)} = \{x_{1,j}, \dots, x_{n,j}\}$. Then in the second sampling scheme we generate $(\theta_j, \mathbf{u}_{(j)})$ as a pair, one margin j at time, from the density

$$f(\theta_j, \mathbf{u}_{(j)} | \theta_{k \neq j}, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}) = f(\theta_j | \theta_{k \neq j}, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}) f(\mathbf{u}_{(j)} | \Theta, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}),$$

by first generating θ_j with $\mathbf{u}_{(j)}$ integrated out, and then $\mathbf{u}_{(j)}$ conditional upon θ_j . The sampling scheme we adopt is therefore:

Sampling Scheme 2 (*One margin at a time*)

(1) For $j = 1, \dots, m$:

(1a) Generate from $f(\theta_j | \theta_{k \neq j}, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x})$

(1b) Generate from $f(\mathbf{u}_{(j)} | \Theta, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x})$

(2) Generate from $f(\phi | \mathbf{u})$

A similar sampler was proposed by Pitt et al. (2006) for the specific case of a Gaussian copula, and this is a generalization to other copula models. In Step (1a) we use a MH step with the same proposal q_j as in Scheme 1, while the conditional posterior of θ_j is

$$\begin{aligned} f(\theta_j | \theta_{k \neq j}, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}) &\propto f(\mathbf{x} | \Theta, \phi, \mathbf{u}_{(k \neq j)}) \pi(\theta_j) \propto \int f(\mathbf{x}, \mathbf{u} | \Theta, \phi) d\mathbf{u}_{(j)} \pi(\theta_j) \\ &= \prod_{i=1}^n \left\{ \int f(\mathbf{x}_i, \mathbf{u}_i | \Theta, \phi) d\mathbf{u}_{i,j} \right\} \pi(\theta_j), \end{aligned}$$

so that from the augmented likelihood in equation (3.1):

$$\begin{aligned} f(\theta_j | \theta_{k \neq j}, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}) &\propto \prod_{i=1}^n \left\{ \int \prod_{k=1}^m \{\mathcal{I}(a_{i,k} \leq u_{i,k} < b_{i,k})\} c(\mathbf{u}_i; \phi) du_{i,j} \right\} \pi(\theta_j) \\ &\propto \prod_{i=1}^n \left\{ \int_{a_{i,j}}^{b_{i,j}} c(\mathbf{u}_i; \phi) du_{i,j} \right\} \pi(\theta_j). \end{aligned}$$

This is a very general expression for any copula. To evaluate the integral it requires the computation of the distribution function $C_{j|k \neq j}$ of the conditional copula:

$$f(\theta_j | \theta_{k \neq j}, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}) \propto \prod_{i=1}^n \{C_{j|k \neq j}(b_{i,j} | u_{i,k \neq j}; \phi) - C_{j|k \neq j}(a_{i,j} | u_{i,k \neq j}; \phi)\} \pi(\theta_j). \quad (3.4)$$

The conditional copula functions above can either be computed in closed form, or numerically, for a wide range of copulas. In Step (1b)

$$\begin{aligned} f(\mathbf{u}_{(j)} | \Theta, \phi, \mathbf{u}_{(k \neq j)}, \mathbf{x}) &\propto f(\mathbf{x} | \Theta, \mathbf{u}) f(\mathbf{u}_{(j)} | \phi, \mathbf{u}_{(k \neq j)}) \\ &\propto \prod_{i=1}^n \mathcal{I}(a_{i,j} \leq u_{i,j} < b_{i,j}) c(\mathbf{u}_i; \phi) \propto \prod_{i=1}^n \mathcal{I}(a_{i,j} \leq u_{i,j} < b_{i,j}) c_{j|k \neq j}(u_{i,j} | u_{i,k \neq j}; \phi). \end{aligned}$$

Therefore, the latents $u_{i,j}$ are generated from the conditional densities $c_{j|k \neq j}$ constrained to $[a_{i,j}, b_{i,j})$, and an iterate for $\mathbf{u}_{(j)}$ obtained.

Last, we make some additional comments regarding the relative merits of the two samplers. First, in Scheme 1 Θ is generated with $\mathbf{u}$ integrated out. While it is tempting to generate Θ conditional upon $\mathbf{u}$ to reduce the computational burden, note that

$$f(\Theta | \phi, \mathbf{u}, \mathbf{x}) \propto f(\mathbf{x} | \Theta, \mathbf{u}) \pi(\Theta) = \prod_{j=1}^m \prod_{i=1}^n \mathcal{I}(a_{i,j} \leq u_{i,j} < b_{i,j}) \pi(\theta_j) = \prod_{j=1}^m f(\theta_j | \mathbf{u}_{(j)}, \mathbf{x}_{(j)}).$$

Because $a_{i,j}$ and $b_{i,j}$ are functions of θ_j , there is likely to be very high dependence between the marginal parameters θ_j and $\mathbf{u}_j$; a similar observation is made by Pitt et al. (2006). Second, for large values of m Scheme 1 is computationally impractical and Scheme 2 preferred. However, for values of m less than about 8, Scheme 1 is our preferred sampler. Third, in much empirical work copula parameters are estimated conditional upon the margins. In this

case Scheme 1 is preferred for all values of m because Θ does not require generation, and $\mathbf{u}$ is generated as a block in a computationally efficient manner. Fourth, $C_{j|1,\dots,j-1}$ needs to be computed to implement Scheme 1, and $C_{j|j \neq k}$ to implement Scheme 2. At least one of these can be computed efficiently for all copula functions that are popular currently. Fifth, throughout we bound $a_{i,j}$ and $b_{i,j}$ to $(\epsilon, 1 - \epsilon)$, with $\epsilon = 0.0001$ to ensure numerical stability.

3.5 Bayesian Estimates

After convergence, K iterates $\{\mathbf{u}^{[k]}, \Theta^{[k]}, \phi^{[k]}; k = 1 \dots, K\}$ are collected from $f(\mathbf{u}, \Theta, \phi|\mathbf{x})$, from which Monte Carlo estimates of the posterior means of parameters are computed and used as point estimates, along with posterior probability intervals.

Dependence measures of $\mathbf{U}$, which has distribution function $C(\mathbf{u}; \phi)$, are functions of ϕ and can be readily estimated. We employ Spearman's pairwise correlation $\rho_{i,j}(\phi) = 12E(U_i U_j) - 3$ for margins i and j . We also employ Kendall's tau $\tau_{i,j}(\phi)$ and the upper and lower tail dependence measures $\lambda_{i,j}^U(\phi) = \lim_{\alpha \uparrow 1} \Pr(U_i > \alpha | U_j > \alpha)$ and $\lambda_{i,j}^L(\phi) = \lim_{\alpha \downarrow 0} \Pr(U_i < \alpha | U_j < \alpha)$. To estimate these based on the fitted copula we compute their expectations with respect to the posterior distribution of the copula parameters $f(\phi|\mathbf{x})$. For example, the estimate of the Spearman correlation is $E(\rho_{i,j}) = \int \rho_{i,j}(\phi) f(\phi|\mathbf{x}) d\phi$. For some copulas, such as elliptical and many Archimedean copulas, the dependence measures can be expressed as a closed form function of ϕ , and the expectations approximated with histogram estimates over the Monte Carlo iterates $\{\phi^{[k]}; k = 1, \dots, K\}$. However, closed form expressions are not readily available for all copulas, including the vine copulas. In these circumstances, we can still estimate the marginal pairwise Spearman's correlation accurately by generating iterates $\mathbf{u}^{[k]} \sim C(\mathbf{u}; \phi^{[k]})$ from the copula at the end of each sweep of the sampling scheme, and then computing $E(\rho_{i,j}) \approx \frac{12}{K} \sum_{k=1}^K u_i^{[k]} u_j^{[k]} - 3$.

Dependence measures for $\mathbf{X}$ with distribution function F in equation (2.1) do not coincide with those for $\mathbf{U}$ when $\mathbf{X}$ is discrete-valued (Denuit & Lambert 2005; Genest & Nešlehová 2007), and are functions of both ϕ and Θ . For example, Kendall's tau between

X_i and X_j (Nešlehová 2007; Genest & Nešlehová 2007) is

$$\begin{aligned} \tau_{i,j}^F(\phi, \Theta) = \sum_{x_i} \sum_{x_j} f_{i,j}(x_i, x_j; \Theta, \phi) [C_{i,j}(F_i, F_j; \phi) + C_{i,j}(F_i, F_j^-; \phi) \\ + C_{i,j}(F_i^-, F_j; \phi) + C_{i,j}(F_i^-, F_j^-; \phi)] - 1, \end{aligned} \quad (3.5)$$

where $C_{i,j}$ is the distribution function of (U_i, U_j) , $f_{i,j}$ is the mass function of (X_i, X_j) , $F_i = F_i(x_i; \theta_i)$ and $F_i^- = F_i(x_i^-; \theta_i)$. This can be estimated by $E(\tau_{i,j}^F) = \int \tau_{i,j}^F(\phi, \Theta) f(\phi, \Theta | \mathbf{x}) d\Theta d\phi$, using the Monte Carlo iterates $(\Theta^{[k]}, \phi^{[k]}) \sim f(\Theta, \phi | \mathbf{x})$ to evaluate the integral. The functions $f_{i,j}$, $C_{i,j}$ and F_j in equation (3.5) can be computed either analytically or numerically from the copula model.

4 Online Retail at Amazon.com

Marketing is an area where copula models with discrete margins have strong potential (Danaher & Smith 2011). To establish the effectiveness of our methodology we first consider two bivariate copula models of consumer behavior at amazon.com, the world's largest online retailer. Because the models are bivariate the Bayesian estimates can be compared with those obtained by maximum likelihood. The data employed were collected by ComScore Inc., and made available by subscription via the Wharton Data Research Service. We analyze a randomly selected sample of $n = 10,000$ visits to amazon.com by US households during 2007. We consider the number of unique page views ($P \in \{1, 2, \dots\}$) and the sales amount ($S \geq 0$) during a visit. Marketing studies treat P as a measure of consumer exposure to a website, and the objective is to measure the level and form of dependence between this and both S and purchase incidence. Website designers hope to observe positive dependence because they try to increase sales by making sites more 'sticky' for visitors; see Danaher (2007).

Table 2 provides a contingency table of the data, aggregated for presentation. Most visits to amazon.com (92.3%) do not result in a sale, so that S is highly zero-inflated. In our first model we treat the margins as fully ordinal-valued and employ empirical distributions for the margins of both S and P . Dependence is captured using Clayton (Clayton 1978), BB7 (Joe 1997; p.153) and Gaussian copulas, which have closed form expressions for the

copula functions and conditional copula distribution functions; see Table 1. The Clayton copula is a single parameter copula with $\lambda^U = 0$, the BB7 is a two parameter copula with asymmetric non-zero tail dependence, unlike the Gaussian copula where $\lambda^U = \lambda^L = 0$. The approach where the margins are estimated in a nonparametric manner, and any dependence captured using a parametric copula estimated in a second step, is widely advocated; see Clayton (1978) for an early example. A second copula model employs a Bernoulli margin for purchase incidence and a negative binomial margin for P (truncated so that $P > 0$), where the latter is a widely used to model exposure counts (Danaher & Smith, 2011). Dependence is again captured using Clayton, BB7 and Gaussian copulas, and in this second model we jointly estimate the parameters of the marginal models and copulas.

—Tables 1 and 3—

Table 3 provides estimates of the copula parameters and some dependence measures for both models. For comparison we also report the maximum likelihood estimates (MLEs), which can be calculated here because the copula is bivariate, and also the pseudo-maximum likelihood estimates (PMLE) obtained by treating the data as continuous. Because the MLE is the posterior mode under flat priors, the Bayesian estimate and MLE are similar, with minor differences due to any asymmetry in the posterior distribution $f(\phi|\mathbf{x})$. However, the PMLE underestimates the level of dependence in the copula, showing that it is important to account for discreteness to obtain accurate likelihood-based estimates.

The level and form of dependence in both models is similar. For the BB7 copula $\hat{\lambda}^U$ is close to zero and $\hat{\lambda}^L = 0.86$ and 0.87 , which is almost identical to that obtained using the Clayton copula, suggesting that the restriction $\lambda^U = 0$ is not unreasonable. Highly asymmetric dependence for the copula suggests that an elliptical copula will fit the dependence structure poorly, with $\hat{\tau} = 0.43$ and 0.44 for the Gaussian copula, which is markedly lower than $\hat{\tau} = 0.70$ and 0.71 for both Archimedean copulas.

From the copula model of (S, P) we also compute estimates of the conditional mean of sales $E[S|P = p] = \sum_s s f_{S,P}(S = s, P = p) / f_P(P = p)$, where the mass function

$$f_{S,P}(S = s, P = p) = C(F_S(s), F_P(p); \phi) - C(F_S(s), F_P(p-1); \phi) \\ + C(F_S(s-1), F_P(p-1); \phi) - C(F_S(s-1), F_P(p); \phi),$$

is evaluated at the posterior mean $\hat{\phi} = E(\phi|\mathbf{x})$. The summation is over the domain of S , but we approximate this over the unique observed values. Figure 1 plots the conditional expectation for values of P between the 2.5th and 97.5th percentiles. For the Archimedean copulas the expected spend in a visit increases as website exposure increases, although at a marginally decreasing rate; the almost linear relationship for the Gaussian copula reflects its more limited dependence structure. For the copula model of P and purchase incidence, the estimates of the marginal parameters (unreported) show that joint estimation with the copula parameters has very little impact on the point estimates, something that is often observed empirically.

Even though both models feature highly discrete margins, for the Gaussian, Clayton and BB7 copulas the proposal g has mean acceptance rates of 72%, 43% and 40% when estimating the first model, and 71%, 48% and 48% when estimating the second. The schemes mix adequately as measured by simulation inefficiency factors (SIFs); see Kim et al (1998) for a discussion of this popular metric. When computed for the parameters in both models using the first 100 autocorrelation coefficients these vary from 5.5 to 134. The largest SIF corresponds to ϕ_2 for the BB7 copula, and SIFs for other parameters are considerably lower.

—Figure 1 about here—

5 D-Vine Copula with Discrete Margins

Vine copula functions C are constructed from a sequence of simpler bivariate copulas called ‘pair-copulas’; see Kurowicka & Cooke (2006), Czado (2010) and Haff et al. (2010) for recent overviews. We consider a D-vine copula, which is well-motivated as a model for longitudinal data, although the approach is equally applicable to other vines. Following Smith et al. (2010) we also extend our Bayesian method to allow for the selection of independence pair-copula components.

5.1 D-vine copula

We outline D-vines here in the context of longitudinal data, where $\mathbf{X}$ in Section 2 has elements ordered in time and distribution function at equation (2.1), but refer the reader to Aas et al. (2009) and Smith et al. (2010) for detailed discussions. A parametric D-vine has a copula density which is the product of $m(m-1)/2$ bivariate copula densities $c_{t,j}$, for $t = 2, \dots, m$ and $j < t$, with

$$c(\mathbf{u}; \phi) = \prod_{t=2}^m c_{t|1, \dots, t-1}(u_t | u_1, \dots, u_{t-1}; \phi) = \prod_{t=2}^m \prod_{j=1}^{t-1} c_{t,j}(u_{t|j+1}, u_{j|t-1}; \phi_{t,j}),$$

where $\mathbf{u} = (u_1, \dots, u_m)$. Each bivariate copula is called a ‘pair-copula’ and has parameters $\phi_{t,j}$. The parameters of the D-vine copula are the collection of all the pair-copula parameters, so that $\phi = \{\phi_{t,s}; t = 2, \dots, m, j < t\}$. The values

$$u_{t|j} = C_{t|j, \dots, t-1}(u_t | u_j, \dots, u_{t-1}; \phi), \quad u_{j|t} = C_{j|j+1, \dots, t}(u_j | u_{j+1}, \dots, u_t; \phi),$$

are computed from $\mathbf{u}$ using a recursive algorithm given in Smith et al. (2010); see also the supplemental material. This involves $m(m-1)$ evaluations of the functions $h_{t,j}(v_1 | v_2; \phi_{t,j}) = \frac{\partial}{\partial v_2} C_{t,j}(v_1, v_2; \phi_{t,j})$, where $C_{t,j}(u_1, u_2; \phi_{t,j}) = \int_0^{u_1} \int_0^{u_2} c_{t,j}(v_1, v_2; \phi_{t,j}) dv_1 dv_2$. That is, $h_{t,j}$ is the conditional distribution function of pair-copula $C_{t,j}$, which is given in closed form in Table 1 for the bivariate copulas employed in this paper.

The conditional distribution function of the D-vine can be expressed as

$$C_{t|1, \dots, t-1}(u_t | u_1, \dots, u_{t-1}; \phi) = h_{t,1} \circ h_{t,2} \circ \dots \circ h_{t,t-1}(u_t),$$

where to evaluate $h_{t,j}(\cdot | u_{j|t-1}; \phi_{t,j})$ for $j = t-1, \dots, 1$, the values $u_{1|t-1}, \dots, u_{t-1|t-1}$ are needed. The inverse

$$C_{t|1, \dots, t-1}^{-1}(\omega_t | u_1, \dots, u_{t-1}; \phi) = h_{t,t-1}^{-1} \circ h_{t,t-2}^{-1} \circ \dots \circ h_{t,1}^{-1}(\omega_t),$$

is used to generate from the D-vine by composition via the inverse distribution method. We

note here that $h_{t,j}^{-1}$ can be evaluated either analytically or numerically as outlined in Table 1.

5.2 Estimation & Pair-Copula Selection

The D-vine can be employed with discrete margins and the posterior distribution evaluated using Scheme 1 as follows. As in Section 3, let $\mathbf{x}_i = (x_{i,1}, \dots, x_{i,m})$ be the i th observation of $\mathbf{X}$, and $\mathbf{u}_i = (u_{i,1}, \dots, u_{i,m})$ be the corresponding latent variable vector. The following algorithm can be used to generate the latent variables from proposal $g(\mathbf{u}_i)$ in Section 3.3:

Algorithm A (*Simulation of Latent Variables for D-Vine*)

For $i = 1, \dots, n$:

(1) Generate $u_{i,1} \sim \text{Uniform}(a_{i,1}, b_{i,1})$

For $j = 2, \dots, m$:

(2) Compute $A_{i,j} = C_{j|1,\dots,j-1}(a_{i,j}|u_{i,1}, \dots, u_{i,j-1}; \phi)$ and $B_{i,j} = C_{j|1,\dots,j-1}(b_{i,j}|u_{i,1}, \dots, u_{i,j-1}; \phi)$; then generate $\omega_{i,j} \sim \text{Uniform}(A_{i,j}, B_{i,j})$

(3) Compute $u_{i,j} = C_{j|1,\dots,j-1}^{-1}(\omega_{i,j}|u_{i,1}, \dots, u_{i,j-1}, \phi)$

(4) Update $u_{i,j|k}$ and $u_{i,k|j}$ values by computing:

(a) $u_{i,j|k} = h_{j,k}(u_{i,j|k+1}|u_{i,k|j-1}; \phi_{j,k})$ for $k = j-1, \dots, 1$

(b) $u_{i,k|j} = h_{j,k}(u_{i,k|j-1}|u_{i,j|k+1}; \phi_{j,k})$ for $k = 1, \dots, j-1$

The values $u_{i,j|k}$ and $u_{i,k|j}$ are the arguments of the pair-copulas for observation $\mathbf{u}_i$, and $\{A_{i,j}, B_{i,j}; j = 2, \dots, m\}$ are also used to evaluate α_i in the MH step.

Smith et al. (2010) also consider selection of independence pair-copulas for continuous margins. Conditional on the latent variables $\mathbf{u} = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$, their method applies without change, thereby extending it to the discrete data case. We summarise the idea here, but refer readers to Smith et al. (2010) for a full exposition. Binary indicator variables $\Gamma = \{\gamma_{t,j}; t = 2, \dots, m; j < t\}$ are introduced to identify whether, or not, each pair-copula is the independence copula, or of a pre-specified pair-copula type c^* . That is, we set

$$c_{t,j}(v_1, v_2; \phi_{t,j}) = \begin{cases} 1 & \text{iff } \gamma_{t,j} = 0 \\ c^*(v_1, v_2; \phi_{t,j}) & \text{iff } \gamma_{t,j} = 1. \end{cases}$$

This specifies a parsimonious inhomogenous Markov process for the longitudinal vector $\mathbf{X} =$

$(X_1, \dots, X_m)$. For example, if $\gamma_{t,j} = 0$ for $j < t - p$, then $c_{t|1,\dots,t-1} = c_{t|t-p,\dots,t-1}$ and $X_t|X_1, \dots, X_{t-1} \sim X_t|X_{t-p}, \dots, X_{t-1}$, so that X_t has Markov order p .

To estimate this model we generate each pair $(\gamma_{t,j}, \phi_{t,j})$, conditional on $\{\Gamma, \phi\} \setminus \{\gamma_{t,j}, \phi_{t,j}\}$ and the latent variables $\mathbf{u}$, using a random walk MH step. We assume the prior $\pi(\Gamma, \phi) = \pi(\Gamma) \prod_{(t,s)} \pi(\phi_{t,s})$, with $\pi(\phi_{t,s})$ differing according to choice of pair-copula, and $\pi(\Gamma)$ chosen to place equal weight on models of different sizes. We note that c^* could easily differ with (t, s) , although we do not consider that here. Also, $h_{t,j}(v_1|v_2, \theta_{t,j}) = v_1$ if $\gamma_{t,j} = 0$, so that when many elements of Γ are 0, Algorithm A is much faster to implement. Overall, the speed of the algorithm is determined by the number of computations required to compute the $h_{t,j}$ functions and their inverses.

5.3 Melbourne Bicycle Path Data

We consider a longitudinal time series of hourly counts of bicycles on an inner city off-road bicycle path in the city of Melbourne, which is part of a transport study by Smith & Kauermann (2011). An induction loop under the path counts the number of bicycles that pass over. The path is mainly used by cyclists who commute to-and-from the central business district during working days. Commuters who use this route have extensive alternative transport options and there is high variation in counts, primarily because commuters switch from cycling to another mode of transport during inclement weather conditions. Data was collected on working days between 12 December 2005 and 19 June 2008, which resulted in $n = 565$ daily observations on hourly counts between 05:01 and 22:00. Figure 2 provides boxplots of the counts for each hourly period, along with plots of counts on three typical days. There are two periods of peak usage, which correspond to the morning commute to work and the late afternoon/early evening return home.

We model the counts during each of the $m = 16$ hourly periods using their empirical distributions. To capture intraday dependence we model the data using three D-vine copulas with Gumbel, Clayton and t-copulas as pair-copulas, along with pair-copula selection. Table 1 outlines these bivariate copulas and their properties. Each Gumbel has an exponential prior on $(\phi_{t,s} - 1)$ with mean 10, and each Clayton an exponential prior on $\phi_{t,s}$ with

mean 10. This places prior weight over a range of values from low to high dependence, as measured by Kendall's tau.² The t-copula is a two parameter copula, with $\phi_{t,s} = \{\psi_{t,s}, \nu_{t,s}\}$, and we adopt an exponential prior for $\nu_{t,s}$ with mean 12 and beta priors for $\psi_{t,s}$ as suggested by Daniels & Pourahmadi (2009). We estimate the parsimonious D-vines using the method outlined, with an initial burnin period of 20,000 sweeps and a Monte Carlo sample of 20,000 iterates. We first discuss the results from the Gumbel and t-copula based vines.

Panels (a) and (d) of Figure 3 plot estimates of the $N = 120$ posterior probabilities $\Pr(\gamma_{t,s} = 1|\mathbf{x})$. Both vines have a high degree of parsimony, although the Gumbel more than the t-copula with $\Pr(\gamma_{t,s} = 1|\mathbf{x}) > 0.25$ for only 28 Gumbel pair-copulas, compared to 84 for the t pair-copulas. The conditional dependence structure of both D-vines indicates strong first order Markov dependence, with $\Pr(\gamma_{t,t-1} = 1|\mathbf{x}) \approx 1$ for $t = 2, \dots, 16$ in both cases. However, what is particularly interesting is the conditional dependence between observations during the morning (hours 1 to 3) and afternoon (hours 11 to 13) peak periods. This is likely due to a 'return trip' effect, where if an individual cycles to work in the morning, then they are much more likely to return by bicycle in the afternoon. Panels (b) and (e) provide the estimates of the posterior means of Kendall's tau $E(\tau_{t,s}|\mathbf{x})$ for the N pair-copulas, showing that this dependence is indeed positive.

—Figures 2 and 3 about here—

The pair-copulas capture conditional dependence, and to measure marginal dependence we compute estimates of the marginal pairwise Spearman's correlations. This is undertaken by simulating iterates $\mathbf{u}^{[k]} \sim C(\mathbf{u}; \phi^{[k]})$ for both D-vines using Algorithm 2 of Smith et al. (2010). Using these iterates we compute the estimates of the Spearman's correlations as discussed in Section 3.5.³ Panels (c) and (f) present the pairwise Spearman correlations of both fitted vines, which are similar and show positive pairwise dependence between counts during the morning and afternoon peaks. An interesting observation is that such extensive dependence arises from two highly parsimonious D-vines.

The same iterates can also be used to estimate other aspects of the fitted distribution.

²Here 95% of the prior weight is on parameter values that correspond to $\tau \in (0.202, 0.973)$ for the Gumbel and $\tau \in (0.112, 0.949)$ for the Clayton.

³Because simulation from a D-vine is fast, we actually simulate 100 iterates from $C(\mathbf{u}; \phi^{[k]})$ for each iterate $\phi^{[k]}$ to reduce the Monte Carlo variation of the expectation.

We construct the bivariate margin in (X_3, X_{12}) , which are the hours with the highest average counts during the morning and afternoon peaks. The fitted distribution function is

$$F_{3,12}(x'_3, x'_{12}) = \int C_{3,12}(F_3(x'_3), F_{12}(x'_{12}); \phi) f(\phi | \mathbf{x}) d\phi, \quad (5.1)$$

where $C_{3,12}$ is the distribution function of (U_3, U_{12}) on $[0, 1]^2$, and is difficult to calculate analytically for a D-vine. Instead, we compute values $x_3^{[k]} = F_3^{-1}(u_3^{[k]})$ and $x_{12}^{[k]} = F_{12}^{-1}(u_{12}^{[k]})$, which are used to construct a bivariate empirical probability mass function. These are given in panels (a) and (b) of Figure 4 for both vines and show the positive, but nonlinear, dependence between the number of cyclists at hours 3 and 12.

—Figure 4 about here.—

To judge the adequacy of all three copulas we compute the fitted values $\hat{x}_{12,i} = E(X_{12} | X_3 = x_{3,i})$ using the fitted distribution at equation (5.1). This corresponds to predicting the number of cyclists in the afternoon peak, given those observed in the morning peak. The mean absolute deviation of the predictions is 32.1, 29.5 and 41.2 for the Gumbel, t and Clayton based D-vines. The mean absolute deviation computed using the sample mean of X_{12} as the prediction is 45.2, suggesting that the Clayton does not capture the dependence structure well. The overall acceptance rates for generating $\mathbf{u}$ were 96.5%, 77.9% and 95.7% for the Gumbel, Clayton and t-copula based vines, suggesting that the MH proposal works well. Last, we note that evaluating $h_{t,j}$ involves many more calculations for the t-copula than the Gumbel or Clayton, and estimation in this case took approximately 48 hours on an older 8 core PC.

6 Mixed Margins

6.1 Data Augmentation

We extend the framework in Section 2 to the case where $\mathbf{X}$ has some discrete and some continuous margins, indexed by $\mathcal{D} = \{j_1, \dots, j_r\}$ and $\mathcal{C} = \{j_{r+1}, \dots, j_m\}$, respectively. We partition $\mathbf{X}$ into the discrete-valued variables $\mathbf{X}_{\mathcal{D}} = \{X_j; j \in \mathcal{D}\}$ and the continuous variables $\mathbf{X}_{\mathcal{C}} = \{X_j; j \in \mathcal{C}\}$; similarly, let $\mathbf{U}_{\mathcal{D}} = \{U_j; j \in \mathcal{D}\}$, $\mathbf{u}_{\mathcal{D}} = \{u_j; j \in \mathcal{D}\}$, $\mathbf{U}_{\mathcal{C}} =$

$\{U_j; j \in \mathcal{C}\}$, and $\mathbf{u}_\mathcal{C} = \{u_j; j \in \mathcal{C}\}$. We assume the same joint density for $(\mathbf{X}, \mathbf{U})$ defined at equation (2.3), but now $f(u_j|x_j) = \mathcal{I}(u_j = F_j(x_j))$ is a point mass for the continuous margins $j \in \mathcal{C}$. The following is a generalization of Proposition 1 to the mixed margin case:

Proposition 3

If $(\mathbf{X}, \mathbf{U})$ has mixed probability density given by equation (2.3), $\mathbf{X}_\mathcal{D}$ are discrete-valued and $\mathbf{X}_\mathcal{C}$ are continuous, then the marginal probability mass function of $\mathbf{X}$ is given by

$$f(\mathbf{x}) = \left(\Delta_{a_{j_1}}^{b_{j_1}} \cdots \Delta_{a_{j_r}}^{b_{j_r}} C_{\mathcal{D}|\mathcal{C}}(v_{j_1}, \dots, v_{j_r} | \mathbf{u}_\mathcal{C}) \right) c(\mathbf{u}_\mathcal{C}) \prod_{j \in \mathcal{C}} f(x_j), \quad (6.1)$$

where $v_{j_1}, \dots, v_{j_r}$ are indices of differencing, $u_j = F(x_j)$ for $j \in \mathcal{C}$, $c(\mathbf{u}_\mathcal{C})$ is the marginal copula density of $\mathbf{U}_\mathcal{C}$ on $[0, 1]^{(m-r)}$ and $\mathbf{u}_\mathcal{C}$ is known exactly given $\mathbf{x}$.

Proof : See Appendix.

The following is a generalization of Proposition 2 and Corollary 1 to the mixed margin case:

Proposition 4

If $(\mathbf{X}, \mathbf{U})$ has mixed probability density given by equation (2.3), $\mathbf{X}_\mathcal{D}$ are discrete-valued and $\mathbf{X}_\mathcal{C}$ are continuous, then

(i) *The density of $\mathbf{U}_\mathcal{D} | \mathbf{X}$ is*

$$f(\mathbf{u}_\mathcal{D} | \mathbf{x}) = \frac{c(\mathbf{u}_\mathcal{C}) \prod_{j \in \mathcal{C}} f(x_j)}{f(\mathbf{x})} c_{\mathcal{D}|\mathcal{C}}(\mathbf{u}_\mathcal{D} | \mathbf{u}_\mathcal{C}) \left(\prod_{j \in \mathcal{D}} \mathcal{I}(a_j \leq u_j < b_j) \right),$$

where $\mathbf{u}_\mathcal{C}$ is known exactly given $\mathbf{x}$.

(ii) *Partition $\mathcal{D}$ into $\mathcal{S}_0 = \{j_1, \dots, j_q\}$ and $\mathcal{S}_1 = \{j_{q+1}, \dots, j_r\}$, denote $\mathbf{u}_{\mathcal{S}_0} = \{u_j; j \in \mathcal{S}_0\}$, $\mathbf{u}_{\mathcal{S}_1} = \{u_j; j \in \mathcal{S}_1\}$ and $\mathbf{U}_{\mathcal{S}_0} = \{U_j; j \in \mathcal{S}_0\}$; then the density of $\mathbf{U}_{\mathcal{S}_0} | \mathbf{X}$ is*

$$\begin{aligned} f(\mathbf{u}_{\mathcal{S}_0} | \mathbf{x}) &= \frac{c(\mathbf{u}_\mathcal{C}) \prod_{j \in \mathcal{C}} f(x_j)}{f(\mathbf{x})} c_{\mathcal{S}_0|\mathcal{C}}(\mathbf{u}_{\mathcal{S}_0} | \mathbf{u}_\mathcal{C}) \\ &\times \Delta_{a_{j_{q+1}}}^{b_{j_{q+1}}} \cdots \Delta_{a_{j_r}}^{b_{j_r}} C_{\mathcal{S}_1|\mathcal{S}_0, \mathcal{C}}(v_{j_{q+1}}, \dots, v_{j_r} | \mathbf{u}_{\mathcal{S}_0}, \mathbf{u}_\mathcal{C}) \left(\prod_{j \in \mathcal{S}_0} \mathcal{I}(a_j \leq u_j < b_j) \right). \end{aligned}$$

(iii) Let $\mathcal{S}_0$ and $\mathcal{S}_1$ be defined as above, and further partition $\mathcal{D}$ into $\mathcal{M}_0 = \{j_1, \dots, j_{q-1}\}$ and $\mathcal{M}_1 = \{j_q, \dots, j_r\}$, with $\mathbf{u}_{\mathcal{M}_0} = \{u_j; j \in \mathcal{M}_0\}$ and $\mathbf{u}_{\mathcal{M}_1} = \{u_j; j \in \mathcal{M}_1\}$, then the density of $U_{j_q}|U_{j_1}, \dots, U_{j_{q-1}}, \mathbf{X}$ is

$$\begin{aligned} f(u_{j_q}|u_{j_1}, \dots, u_{j_{q-1}}, \mathbf{x}) &= c_{j_q|\mathcal{M}_0, \mathcal{C}}(u_{j_q}|\mathbf{u}_{\mathcal{M}_0}, \mathbf{u}_{\mathcal{C}})\mathcal{I}(a_{j_q} \leq u_{j_q} < b_{j_q}) \\ &\times \frac{\Delta_{a_{j_q+1}}^{b_{j_q+1}} \dots \Delta_{a_{j_r}}^{b_{j_r}} C_{\mathcal{S}_1|\mathcal{S}_0, \mathcal{C}}(v_{j_{q+1}}, \dots, v_{j_r}|\mathbf{u}_{\mathcal{S}_0}, \mathbf{u}_{\mathcal{C}})}{\Delta_{a_{j_q}}^{b_{j_q}} \dots \Delta_{a_{j_r}}^{b_{j_r}} C_{\mathcal{M}_1|\mathcal{M}_0, \mathcal{C}}(v_{j_q}, \dots, v_{j_r}|\mathbf{u}_{\mathcal{M}_0}, \mathbf{u}_{\mathcal{C}})}. \end{aligned}$$

Proof : See Appendix.

6.2 Bayesian Estimation

The two sampling schemes outlined in Section 3 can be used to estimate the copula model with the following modifications. First, in both schemes $\mathbf{u}_{(j)}$ is not generated for $j \in \mathcal{C}$ because $u_{i,j} = F_j(x_{i,j}; \theta_j) = b_{i,j} = a_{i,j}$. Second, the MH step in Section 3.3 is used to generate $\mathbf{u}_{i,\mathcal{D}} = \{u_{i,j_q}; j_q \in \mathcal{D}\}$ with proposal $g(\mathbf{u}_{i,\mathcal{D}}) = g_{j_1}(u_{i,j_1}|\mathbf{u}_{\mathcal{C}}) \prod_{q=2}^r g_{j_q}(u_{i,j_q}|\mathbf{u}_{i,\mathcal{M}_0}, \mathbf{u}_{\mathcal{C}})$, where we denote $\mathbf{u}_{i,\mathcal{W}} = \{u_{i,j}; j \in \mathcal{W}\}$. Each density g_{j_q} is proportional to the conditional copula $c_{j_q|\mathcal{M}_0, \mathcal{C}}$, truncated to $[a_{i,j_q}, b_{i,j_q})$, while

$$\alpha_i = \prod_{j_q \in \mathcal{D}} \frac{C_{j_q|\mathcal{M}_0, \mathcal{C}}(b_{i,j_q}|\mathbf{u}_{i,\mathcal{M}_0}^{\text{new}}, \mathbf{u}_{i,\mathcal{C}}; \phi) - C_{j_q|\mathcal{M}_0, \mathcal{C}}(a_{i,j_q}|\mathbf{u}_{i,\mathcal{M}_0}^{\text{new}}, \mathbf{u}_{i,\mathcal{C}}; \phi)}{C_{j_q|\mathcal{M}_0, \mathcal{C}}(b_{i,j_q}|\mathbf{u}_{i,\mathcal{M}_0}^{\text{old}}, \mathbf{u}_{i,\mathcal{C}}; \phi) - C_{j_q|\mathcal{M}_0, \mathcal{C}}(a_{i,j_q}|\mathbf{u}_{i,\mathcal{M}_0}^{\text{old}}, \mathbf{u}_{i,\mathcal{C}}; \phi)}.$$

The copula parameters are generated conditional upon $\mathbf{u}$ as before. To generate the marginal parameters in Scheme 1 using equation (3.3), the likelihood $f(\mathbf{x}|\Theta, \phi)$ is replaced with that at equation (6.1). Last, when $j \in \mathcal{C}$ the form of the marginal posterior at Step 1(a) of Scheme 2 differs from that in equation (3.4), and is instead

$$f(\theta_j|\theta_{k \neq j}, \phi, \mathbf{u}_{k \neq j}, \mathbf{x}) \propto \pi(\theta_j) \prod_{i=1}^n f(x_{i,j}|\theta_j) c(\mathbf{u}_i; \phi), \text{ for } j \in \mathcal{C},$$

where the last term cannot be dropped because $u_{i,j}$ is a function of θ_j .

To illustrate we fit a copula model to a subset of the online retail data comprising the $n = 768$ purchases at amazon.com. A log-normal margin is used for $S > 0$, and a negative

binomial truncated to be positive for $P \in \{1, 2, \dots\}$, along with a BB7 copula. There remained some small, but significant, positive dependence with $\hat{\tau} = 0.116$ and a 95% posterior probability interval of (0.065, 0.170). Both the lower and upper tail dependence was close to zero. These results suggest that dependence between page views and sales is mostly due to purchase incidence, rather than amount.

7 Discussion

Many existing parametric models for discrete data can be expressed as copula models; for example, a multivariate probit model can be expressed as a Gaussian copula model (Song 2000). Our method therefore extends the popular data augmentation approach of Chib and Greenberg (1998) for the multivariate probit to a much wider class of models. We also note that analysis using other augmented likelihoods is possible, and that the density at equation (2.3) is only one choice. In particular, for copulas constructed from a multivariate distribution G by inversion of Sklar's theorem (Nelsen 2006, Sect. 3.1) it is attractive to consider augmentation with variables distributed as G , as for the Gaussian copula in Pitt et al. (2006) and skew t copula in Smith, Gan & Kohn (2010). However, our approach is more general and applies to any copula as long as the conditional copula distribution functions can be evaluated. This is particularly useful for many copulas currently in use, such as Archimedean and vine copulas, where it would be hard to envisage a more appropriate augmentation than that at equation (2.3). That such non-elliptical copulas are required to capture dependence in some multivariate data is demonstrated in our empirical work.

Denuit & Lambert (2005) show that pairwise concordance is unaffected by subtracting independent uniforms from bivariate discrete data. Madsen & Fang (2010) use this to specify an approach to maximise the likelihood of a Gaussian copula model using unconstrained independent latent uniforms. This is completely different from the data augmentation we suggest. In our approach the latents are jointly distributed by the copula function and are truncated, conditional upon the data. Genest & Nešlehová (2007) highlight the importance of computing estimates of discrete-margined parametric copula models using full likelihood-based methods. The data augmentation approach we suggest provides a general and com-

putationally viable avenue to compute such inference. The Bayesian framework also allows for the adoption of informative priors, such as shrinkage priors in more highly-parameterized models, or for point mass priors to enable Bayesian selection and model averaging. We show the latter provides insights on the dependence structure of longitudinal count data for a 16 dimensional D-vine in Section 5.

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Supplemental Materials

These contain the algorithm used to evaluate the arguments of the D-vine in Section 5.1, and a note on the equivalence of the notation used here and in Smith et al. (2010). The Melbourne bicycle path data are also included.

Appendix

This appendix provides the proofs of the propositions found in Sections 2 and 6.

Proof of Proposition 1

To show this, integrate over $\mathbf{u}$:

$$f(\mathbf{x}) = \int f(\mathbf{x}, \mathbf{u}) d\mathbf{u} = \int c(\mathbf{u}) \prod_{j=1}^m \mathcal{I}(F_j(x_j^-) \leq u_j < F_j(x_j)) d\mathbf{u} = \Delta_{a_1}^{b_1} \cdots \Delta_{a_m}^{b_m} C(\mathbf{v}).$$

Proof of Proposition 2

First, the joint distribution of $\mathbf{U}$ can be written as

$$\begin{aligned} c(\mathbf{u}) &= \frac{\partial^{m-j}}{\partial u_{j+1} \cdots \partial u_m} \left(\frac{\partial^j}{\partial u_1 \cdots \partial u_j} C(\mathbf{u}) \right) \\ &= \frac{\partial^{m-j}}{\partial u_{j+1} \cdots \partial u_m} \left\{ c(u_1, \dots, u_j) C_{j+1, \dots, m|1, \dots, j}(u_{j+1}, \dots, u_m | u_1, \dots, u_j) \right\}. \end{aligned}$$

Then, from equation (2.4):

$$\begin{aligned} f(u_1, \dots, u_j | \mathbf{x}) &= \int \cdots \int f(\mathbf{u} | \mathbf{x}) du_{j+1} \cdots du_m \\ &= \frac{\prod_{k=1}^j \mathcal{I}(a_k \leq u_k < b_k)}{f(\mathbf{x})} \int_{a_{j+1}}^{b_{j+1}} \cdots \int_{a_m}^{b_m} c(\mathbf{u}) du_{j+1} \cdots du_m \\ &= \frac{\prod_{k=1}^j \mathcal{I}(a_k \leq u_k < b_k)}{f(\mathbf{x})} c(u_1, \dots, u_j) \Delta_{a_{j+1}}^{b_{j+1}} \cdots \Delta_{a_m}^{b_m} C_{j+1, \dots, m|1, \dots, j}(u_{j+1}, \dots, u_m | u_1, \dots, u_j). \end{aligned}$$

To prove Propositions 3 and 4, we use the following identity that can be derived using standard measure theory; see Stein & Shakarchi (2005) or Schilling (2005). Let $H_1, \dots, H_k$ be the distribution functions of absolutely continuous real random variables, with density functions $h_j(x_j) = dH_j(x_j)/dx_j$, for $j = 1, \dots, k$. Then, for g any measurable function:

$$\int_0^1 \cdots \int_0^1 \left(\prod_{j=1}^k \mathcal{I}(u_j = H_j(x_j)) \right) g(u_1, \dots, u_k) du_1 \cdots du_k = g(H_1(x_1), \dots, H_k(x_k)) \prod_{j=1}^k h_j(x_j).$$

Proof of Proposition 3

First note that because $f(x_j|u_j) = \mathcal{I}(u_j = F_j(X_j))$ for $j \in \mathcal{C}$ then

$$\begin{aligned} f(\mathbf{u}, \mathbf{x}) &= c(\mathbf{u}) \prod_{j=1}^m f(x_j|u_j) \\ &= c_{\mathcal{D}|\mathcal{C}}(\mathbf{u}_{\mathcal{D}}|\mathbf{u}_{\mathcal{C}})c(\mathbf{u}_{\mathcal{C}}) \prod_{j \in \mathcal{D}} \mathcal{I}(F_j(x_j^-) \leq u_j < F_j(x_j)) \prod_{j \in \mathcal{C}} \mathcal{I}(u_j = F_j(x_j)). \end{aligned}$$

The marginal probability mass function is therefore

$$\begin{aligned} f(\mathbf{x}) &= \int_{[0,1]^r} \prod_{j \in \mathcal{D}} \mathcal{I}(a_j \leq \tilde{u}_j < b_j) \left\{ \int_{[0,1]^{m-r}} c_{\mathcal{D}|\mathcal{C}}(\tilde{\mathbf{u}}_{\mathcal{D}}|\tilde{\mathbf{u}}_{\mathcal{C}})c(\tilde{\mathbf{u}}_{\mathcal{C}}) \prod_{j \in \mathcal{C}} \mathcal{I}(\tilde{u}_j = F_j(x_j)) d\tilde{\mathbf{u}}_{\mathcal{C}} \right\} d\tilde{\mathbf{u}}_{\mathcal{D}} \\ &= \left\{ \int_{[0,1]^r} \prod_{j \in \mathcal{D}} \mathcal{I}(a_j \leq \tilde{u}_j < b_j) c_{\mathcal{D}|\mathcal{C}}(\tilde{\mathbf{u}}_{\mathcal{D}}|\mathbf{u}_{\mathcal{C}}) d\tilde{\mathbf{u}}_{\mathcal{D}} \right\} c(\mathbf{u}_{\mathcal{C}}) \prod_{j \in \mathcal{C}} f_j(u_j) \\ &= \Delta_{a_{j_1}}^{b_{j_1}} \cdots \Delta_{a_{j_r}}^{b_{j_r}} C_{\mathcal{D}|\mathcal{C}}(v_{j_1}, \dots, v_{j_r}|\mathbf{u}_{\mathcal{C}})c(\mathbf{u}_{\mathcal{C}}) \prod_{j \in \mathcal{C}} f_j(u_j), \end{aligned}$$

where $u_j = F_j(x_j)$ for $j \in \mathcal{C}$, and $b_j = F_j(x_j)$, $a_j = F_j(x_j^-)$ for $j \in \mathcal{D}$.

Proof of Proposition 4

Part (i): Note that

$$\begin{aligned} f(\mathbf{u}|\mathbf{x}) &= \frac{c(\mathbf{u}) \prod_{j=1}^m f(x_j|u_j)}{f(\mathbf{x})} \\ &= \frac{c_{\mathcal{D}|\mathcal{C}}(\mathbf{u}_{\mathcal{D}}|\mathbf{u}_{\mathcal{C}})c(\mathbf{u}_{\mathcal{C}})}{f(\mathbf{x})} \prod_{j \in \mathcal{D}} \mathcal{I}(F_j(x_j^-) \leq u_j < F_j(x_j)) \prod_{j \in \mathcal{C}} \mathcal{I}(u_j = F_j(x_j)). \end{aligned}$$

Therefore, the margin in $\mathbf{u}_{\mathcal{D}}$ is

$$\begin{aligned} f(\mathbf{u}_{\mathcal{D}}|\mathbf{x}) &= \int_{[0,1]^{m-r}} c_{\mathcal{D}|\mathcal{C}}(\mathbf{u}_{\mathcal{D}}|\tilde{\mathbf{u}}_{\mathcal{C}})c(\tilde{\mathbf{u}}_{\mathcal{C}}) \prod_{j \in \mathcal{C}} \mathcal{I}(\tilde{u}_j = F_j(x_j)) d\tilde{\mathbf{u}}_{\mathcal{C}} \frac{\prod_{j \in \mathcal{D}} \mathcal{I}(a_j \leq u_j < b_j)}{f(\mathbf{x})} \\ &= \frac{c(\mathbf{u}_{\mathcal{C}}) \prod_{j \in \mathcal{C}} f(x_j)}{f(\mathbf{x})} c_{\mathcal{D}|\mathcal{C}}(\mathbf{u}_{\mathcal{D}}|\mathbf{u}_{\mathcal{C}}) \left(\prod_{j \in \mathcal{D}} \mathcal{I}(a_j \leq u_j < b_j) \right). \end{aligned}$$

Part (ii): The result follows from integrating $\mathbf{u}_{\mathcal{S}_1}$ out of $f(\mathbf{u}_{\mathcal{D}}|\mathbf{x})$.

Part (iii): This follows from part (ii) and the application of Bayes theorem.

References

- Aas, K., C. Czado, A. Frigessi & H. Bakken (2009), 'Pair-copula constructions of multiple dependence', *Insurance: Mathematics and Economics*, 44, 182-198.
- Bedford, T. & R. Cooke (2002), 'Vines - a new graphical model for dependent random variables', *Annals of Statistics*, 30, 1031-1068.
- Cameron, A., L. Tong, P. Trivedi & D. Zimmer (2004), 'Modelling the differences in counted outcomes using bivariate copula models with application to mismeasured counts', *Econometrics Journal*, 7, 566-584.
- Cherubini, U., E. Luciano & W. Vecchiato (2004), *Copula methods in finance*, New York, NY: Wiley.
- Chib, S. & E. Greenberg (1998), 'Analysis of multivariate probit models', *Biometrika*, 85, 347-361.
- Clayton, D. (1978), 'A model for association in bivariate life tables and its application to epidemiological studies of family tendency in chronic disease incidence', *Biometrika*, 65, 141-151.
- Czado, C. (2010), 'Pair-copula constructions of multivariate copulas', In P. Jaworski, F. Durante, W. Härdle, and T. Rychlik (Eds.), *Copula Theory and Its Applications*, Berlin: Springer.
- Danaher, P. (2007) 'Modeling page views across multiple websites with an application to internet reach and frequency prediction', *Marketing Science*, 26, 422-437.
- Danaher, P., & M. Smith (2011), 'Modeling multivariate distributions using copulas: applications in marketing' (with discussion), *Marketing Science*, 30, 4-21.
- Daniels, M. & M. Pourahmadi (2009), 'Modeling covariance matrices via partial autocorrelations', *Journal of Multivariate Analysis*, 100, 2352-2363.
- Denuit, M. & P. Lambert (2005), 'Constraints on concordance measures in bivariate discrete data', *Journal of Multivariate Analysis*, 93, 40-57.
- Frees, E. & E. Valdez (1998), 'Understanding relationships using copulas', *North American Actuarial Journal*, 2, 1-25.
- Genest, C., K. Ghoudi & L. P. Rivest (1995) 'A semiparametric estimation procedure of dependence parameters in multivariate families of distributions', *Biometrika*, 82, 543-552.
- Genest, C. & J. Nešlehová (2007) 'A primer on copulas for count data', *The Astin Bulletin*, 37, 475-515.
- Haff, I., K. Aas & A. Frigessi (2010), 'On the simplified pair-copula construction- simply useful or too simplistic?', *Journal of Multivariate Analysis*, 101, 1296-1310.

- Joe, H. (1996), 'Families of m -variate distributions with given margins and $m(m - 1)/2$ bivariate dependence parameters', In: L. Rüschendorf, B. Schweizer & M. Taylor, (Eds.), *Distributions with Fixed Marginals and Related Topics*.
- Joe, H. (1997), *Multivariate Models and Dependence Concepts*, Chapman & Hall.
- Jondeau, E. & M. Rockinger (2006), 'The Copula-GARCH model of conditional dependencies: An international stock market application', *Journal of International Money and Finance*, 25, 827-853.
- Kim, S., N. Shephard & S. Chib (1998), 'Stochastic volatility: likelihood inference and comparison with ARCH models', *Review of Economic Studies*, 65, 361-393.
- Kurowicka, D., & R. M. Cooke (2006), *Uncertainty Analysis with High Dimensional Dependence Modelling*, Wiley: New York.
- Liang, K.Y., S. Zeger & B. Qaqish (1992), 'Multivariate regression analyses for categorical data', *Journal of the Royal Statistical Society, Series B*, 54, 3-40.
- Madsen, L. and Y. Fang (2010), 'Joint regression analysis for discrete longitudinal data', *Biometrics* (with comment), to appear.
- McNeil, A. J., R. Frey & R. Embrechts (2005), *Quantitative Risk Management: Concepts, Techniques and Tools*, Princeton University Press, Princeton: NJ.
- Min, A. & C. Czado (2010), 'Bayesian inference for multivariate copulas using pair-copula constructions', *Journal of Financial Econometrics*, 8, 511-546.
- Nelsen, R. B. (2006), *An Introduction to Copulas*, 2nd ed., Springer
- Nešlehová, J. (2007), 'On rank correlation measures for non-continuous random variables', *Journal of Multivariate Analysis*, 98, 544-567.
- Nikoloulopoulos, A. & D. Karlis (2008), 'Multivariate logit copula model with an application to dental data', *Statistics in Medicine*, 27, 6393-6406.
- Oakes, D., (1989), 'Bivariate survival models induced by frailties', *Journal of the American Statistical Association*, 84, 487-493.
- Patton, A.J. (2006), 'Modelling asymmetric exchange rate dependence', *International Economic Review*, 47, 527-556.
- Pitt, M., D. Chan & R. Kohn (2006), 'Efficient Bayesian inference for Gaussian copula regression models', *Biometrika*, 93, 537-554.
- Robert, C. & G. Casella (2004), *Monte Carlo Statistical Methods*, (2nd ed.), New York, NY: Springer.
- Schilling, R. L. (2005), *Measures, Integrals and Martingales*, Cambridge University Press.

Sklar, A. (1959), 'Fonctions de répartition à n dimensions et leurs marges', *Publications de l'Institut de Statistique de L'Université de Paris*, 8, 229-231.

Smith, M., Q. Gan & R. Kohn (2010), 'Modelling dependence using skew t copulas: Bayesian inference and applications', *Journal of Applied Econometrics*, (forthcoming), DOI: 10.1002/jae.1215.

Smith, M. S. & G. Kauermann (2011), 'Bicycle Commuting in Melbourne during the 2000s Energy Crisis: A Semiparametric Analysis of Intraday Volumes', *Journal of Transportation Research Part B*, (forthcoming).

Smith, M., A. Min, C. Almeida & C. Czado (2010), 'Modeling longitudinal data using a pair-copula decomposition of serial dependence', *Journal of the American Statistical Association*, 105, 1467-1479.

Song, P., M. Li and Y. Yuan (2009), 'Joint regression analysis of correlated data using Gaussian copulas', *Biometrics*, 65, 60-68.

Stein, E. M. & R. Shakarchi (2005), *Princeton Lectures on Analysis III Real Analysis: Measure Theory, Integration and Hilbert Spaces*, Princeton University Press.

Song, P. (2000), 'Multivariate dispersion models denenerated from Gaussian copula', *Scandinavian Journal of Statistics*, 27, 305-320.

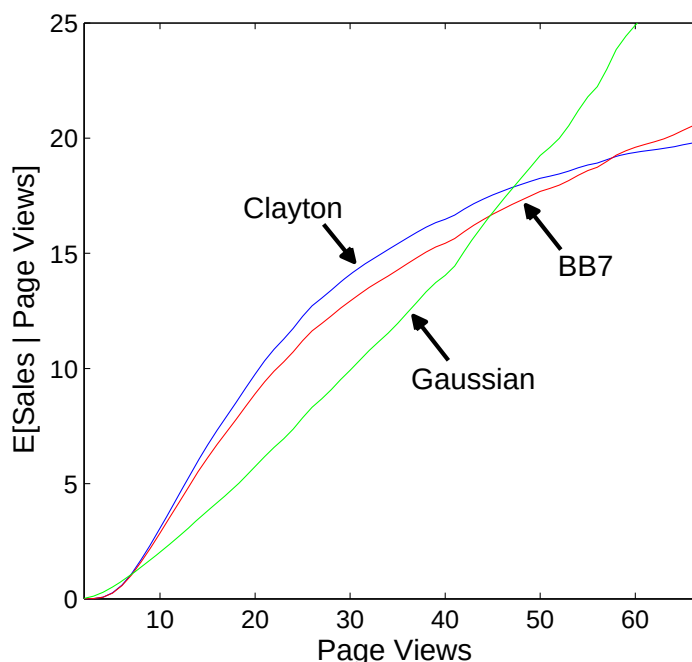


Figure 1: The expected value of sales (S), conditional upon the number of page views (P), for amazon.com using three different copulas. The expectation is plotted between the 2.5% and 97.5% percentiles of the observed values of P .

Clayton ($\phi \in (-1, \infty) \setminus \{0\}$)
$C(u_1, u_2; \phi) = \max \left\{ (u_1^{-\phi} + u_2^{-\phi} - 1)^{-1/\phi}, 0 \right\}$
$C_{1 2}(u_1 u_2; \phi) = \max \left\{ u_2^{-(1+\phi)} \left(u_1^{-\phi} + u_2^{-\phi} - 1 \right)^{-(1+1/\phi)}, 0 \right\}$
$C_{1 2}^{-1}(v u_2; \phi) = \left(1 - u_2^{-\phi} + \left[v u_2^{(1+\phi)} \right]^{-\phi/(\phi+1)} \right)^{-1/\phi}$
$\tau_{1,2}(\phi) = \phi/(\phi+2)$, $\lambda_{1,2}^L(\phi) = 2^{-1/\phi}$ and $\lambda_{1,2}^U(\phi) = 0$
Gumbel ($\phi \geq 1$)
$C(u_1, u_2; \phi) = \exp(-(\tilde{u}_1^\phi + \tilde{u}_2^\phi)^{1/\phi})$, where $\tilde{u}_j = -\log(u_j)$
$C_{1 2}(u_1 u_2; \phi) = C(u_1, u_2; \phi) \frac{1}{u_2} (\tilde{u}_2)^{\phi-1} \left[\tilde{u}_1^\phi + \tilde{u}_2^\phi \right]^{1/\phi-1}$
$C_{1 2}^{-1}$: Obtained Numerically using Newton's Method
$\tau_{1,2}(\phi) = 1 - \phi^{-1}$, $\lambda_{1,2}^L(\phi) = 0$ and $\lambda_{1,2}^U(\phi) = 2 - 2^{1/\phi}$
BB7 ($\phi = (\phi_1, \phi_2)$ with $\phi_1 \geq 1$ and $\phi_2 > 0$)
$C(u_1, u_2; \phi) = 1 - \left(1 - \left[(1 - \bar{u}_1^{\phi_1})^{-\phi_2} + (1 - \bar{u}_2^{\phi_1})^{-\phi_2} - 1 \right]^{-1/\phi_2} \right)^{1/\phi_1}$
where $\bar{u}_j = 1 - u_j$
$C_{1 2}(u_1 u_2; \phi) = (1 - \omega^{-1/\phi_2})^{(1/\phi_1-1)} \omega^{-(1/\phi_2+1)} (1 - \bar{u}_2^{\phi_1})^{-(\phi_2+1)} \bar{u}_2^{(\phi_1-1)}$
where $\omega = (1 - \bar{u}_1^{\phi_1})^{-\phi_2} + (1 - \bar{u}_2^{\phi_1})^{-\phi_2} - 1$
$C_{1 2}^{-1}$: Obtained Numerically using Newton's Method
$\tau_{1,2}(\phi) = 1 - \frac{4}{\phi_1^2 \phi_2} \left[B\left(2, \frac{2}{\phi_1} - 1\right) - B\left(\phi_2 + 2, \frac{2}{\phi_1} - 1\right) \right]$ for $0 \leq \phi_1 < 2$ only
$\lambda_{1,2}^L(\phi) = 2^{-1/\phi_2}$ and $\lambda_{1,2}^U(\phi) = 2 - 2^{1/\phi_1}$
Bivariate t-copula ($\phi = (\psi, \nu)$ with $-1 < \psi < 1$ and $\nu > 0$)
$C(u_1, u_2; \phi) = T_\nu(t_\nu^{-1}(u_1), t_\nu^{-1}(u_2); \psi)$
$C_{1 2}(u_1 u_2; \phi) = t_{\nu+1} \left(\left[\frac{\nu+1}{\nu+(t_\nu^{-1}(u_2))^2} \right]^{1/2} \frac{t_\nu^{-1}(u_1) - \psi t_\nu^{-1}(u_2)}{\sqrt{1-\psi^2}} \right)$
$C_{1 2}^{-1}(v u_2; \phi) = t_\nu \left(\left[\frac{(1-\psi^2)(\nu+(t_\nu^{-1}(u_2))^2)}{\nu+1} \right]^{1/2} t_{\nu+1}^{-1}(v) + \psi t_\nu^{-1}(u_2) \right)$
$\rho_{1,2}(\phi) = \frac{6}{\pi} \arcsin \frac{\psi}{2}$ and $\tau_{1,2}(\phi) = \frac{2}{\pi} \arcsin \psi$
$\lambda_{1,2}^L(\phi) = \lambda_{1,2}^U(\phi) = 2t_{\nu+1} \left(-\sqrt{\frac{(\nu+1)(1-\psi)}{1+\psi}} \right)$

Table 1: Copula functions, dependence measures, conditional distribution and density functions for four bivariate copulas. The Clayton and Gumbel copulas are single parameter copulas, while the BB7 and t are two parameter copulas. For the BB7 copula, $B(\cdot, \cdot)$ is the Beta function, and for the bivariate t-copula the function t_ν is the standard t distribution function and T_ν is the bivariate t distribution function with correlation ψ . The Gaussian copula is outlined in detail in Song (2000).

				<i>Page Views</i>				
<i>Sales</i>	1-5	6-10	11-20	21-30	31-40	≥ 41	Total	
$S = \$0$	4070	2342	1568	550	240	462	9232	
$\$0 < S \leq \15	1	16	57	33	16	34	157	
$\$15 < S \leq \30	2	32	67	39	20	52	212	
$\$30 < S \leq \45	2	11	39	43	26	46	167	
$\$45 < S \leq \70	0	6	31	22	15	40	114	
$\$S > \70	1	8	24	21	17	47	118	
Total	4076	2415	1786	708	334	681	10000	

Table 2: Contingency table for Sales (S) and Page views (P) of a sub-sample of online visits by US households to amazon.com during 2007. The data have been aggregated to ranges for presentation purposes only.

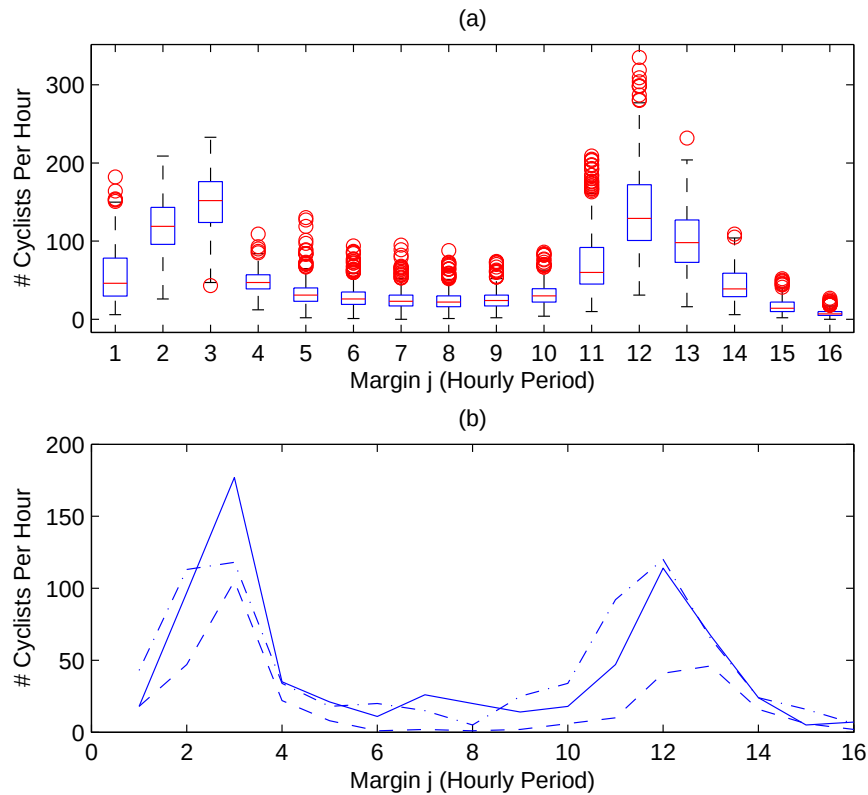


Figure 2: Panel (a): Boxplots of the hourly counts of the number of cyclists passing over the induction loop on the Melbourne bike path. Panel (b): Plots of hourly counts for three randomly selected days in the sample. In both panels the data is broken down by hour of day, with X_1 being the count between 05:01 and 06:00, and X_{16} the count between 21:01 and 22:00.

	<i>Model 1: Sales Amount</i>			<i>Model 2: Purchase Incidence</i>		
	<i>Bayes</i>	<i>MLE</i>	<i>PMLE</i>	<i>Bayes</i>	<i>MLE</i>	<i>PMLE</i>
<i>Clayton Copula</i>						
$\hat{\phi}$	4.635 (4.247, 5.047)	4.679 (0.206)	0.246 (0.014)	4.960 (4.616, 5.309)	5.099 (0.182)	0.838 (0.020)
$\hat{\tau}$	0.698 (0.680, 0.716)	0.701 (0.009)	0.110 (0.005)	0.713 (0.698, 0.726)	0.718 (0.007)	0.293 (0.005)
$\hat{\lambda}^L$	0.861 (0.849, 0.872)	0.862 (0.006)	0.060 (0.009)	0.869 (0.861, 0.878)	0.873 (0.004)	0.437 (0.009)
$\hat{\tau}^F$	0.1031 (0.1004, 0.1055)	0.1031 (0.0014)	— —	0.1056 (0.1037, 0.1072)	0.1055 (0.0010)	— —
<i>BB7 Copula</i>						
$\hat{\phi}_1$	1.048 (1.015, 1.093)	1.043 (0.020)	1.013 (0.004)	1.008 (1.000, 1.026)	1.000 (0.030)	1.000 (0.001)
$\hat{\phi}_2$	4.631 (4.172, 5.046)	4.590 (0.216)	0.229 (0.015)	4.972 (4.589, 5.308)	5.095 (0.183)	0.837 (0.020)
$\hat{\tau}$	0.697 (0.675, 0.715)	0.695 (0.010)	0.109 (0.005)	0.713 (0.696, 0.726)	0.718 (0.007)	0.295 (0.005)
$\hat{\lambda}^L$	0.861 (0.847, 0.872)	0.860 (0.006)	0.048 (0.009)	0.870 (0.860, 0.878)	0.873 (0.004)	0.440 (0.009)
$\hat{\lambda}^U$	0.062 (0.020, 0.115)	0.056 (0.025)	0.018 (0.006)	0.011 (0.000, 0.034)	0.000 (0.041)	0.000 (0.001)
$\hat{\tau}^F$	0.1039 (0.1010, 0.1065)	0.1033 (0.0014)	— —	0.1048 (0.1042, 0.1055)	0.1055 (0.0013)	— —
<i>Gaussian Copula</i>						
$\hat{\phi}$	0.622 (0.600, 0.644)	0.624 (0.012)	0.112 (0.007)	0.635 (0.506, 0.738)	0.637 (0.068)	0.128 (0.027)
$\hat{\tau}$	0.428 (0.410, 0.445)	0.429 (0.010)	0.072 (0.005)	0.440 (0.337, 0.528)	0.440 (0.056)	0.081 (0.017)
$\hat{\tau}^F$	0.0978 (0.0945, 0.1011)	0.0983 (0.0017)	— —	0.0983 (0.0806, 0.1128)	0.0990 (0.0096)	— —

Table 3: Parameter estimates for the Clayton, BB7 and Gaussian bivariate copulas for the copula models of sales amount and purchase incidence at amazon.com. Also included are the estimates of Kendall's tau ($\hat{\tau}$) and the lower and upper tail dependence indices $\hat{\lambda}^L$ and $\hat{\lambda}^U$ for the fitted copula functions. The estimates of Kendall's tau ($\hat{\tau}^F$) for the discrete data at equation (3.5) are also provided for the two discrete copula models, although this metric is hard to interpret; see Genest & Nešlehová (2007, p.492). The 95% posterior probability intervals for the Bayesian estimates, and standard errors for the maximum likelihood based estimates, are given in parentheses.

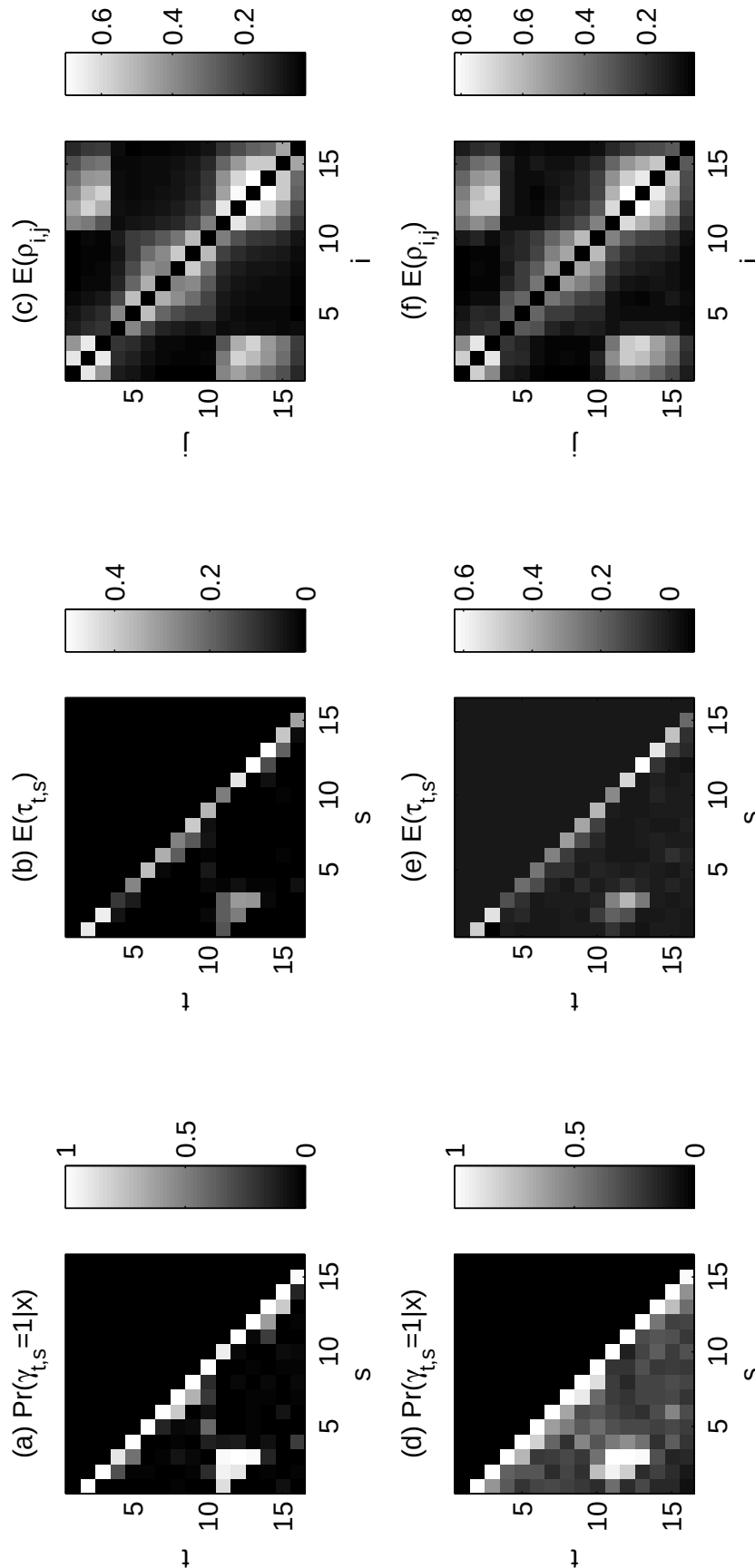


Figure 3: Estimates from two D-vines fit to the Melbourne bicycle path data. The upper row corresponds to results when Gumbel pair-copulas are used, and the lower row when t pair-copulas are used. Panels (a) and (d) provide the posterior probabilities $\Pr(\gamma_{t,s}=1|\mathbf{x})$, for $s < t$ and $t = 2, \dots, 16$, that each pair-copula is not the independence copula in the bottom triangle. Panels (b) and (e) provide the estimate of Kendall's tau $E(\tau_{t,s})$ for each pair-copula $c_{t,s}$ from the two fitted vines. Panels (c) and (f) are the estimates of the marginal pairwise Spearman's correlations $E(\rho_{i,j})$, for all i, j , from the fitted vines.

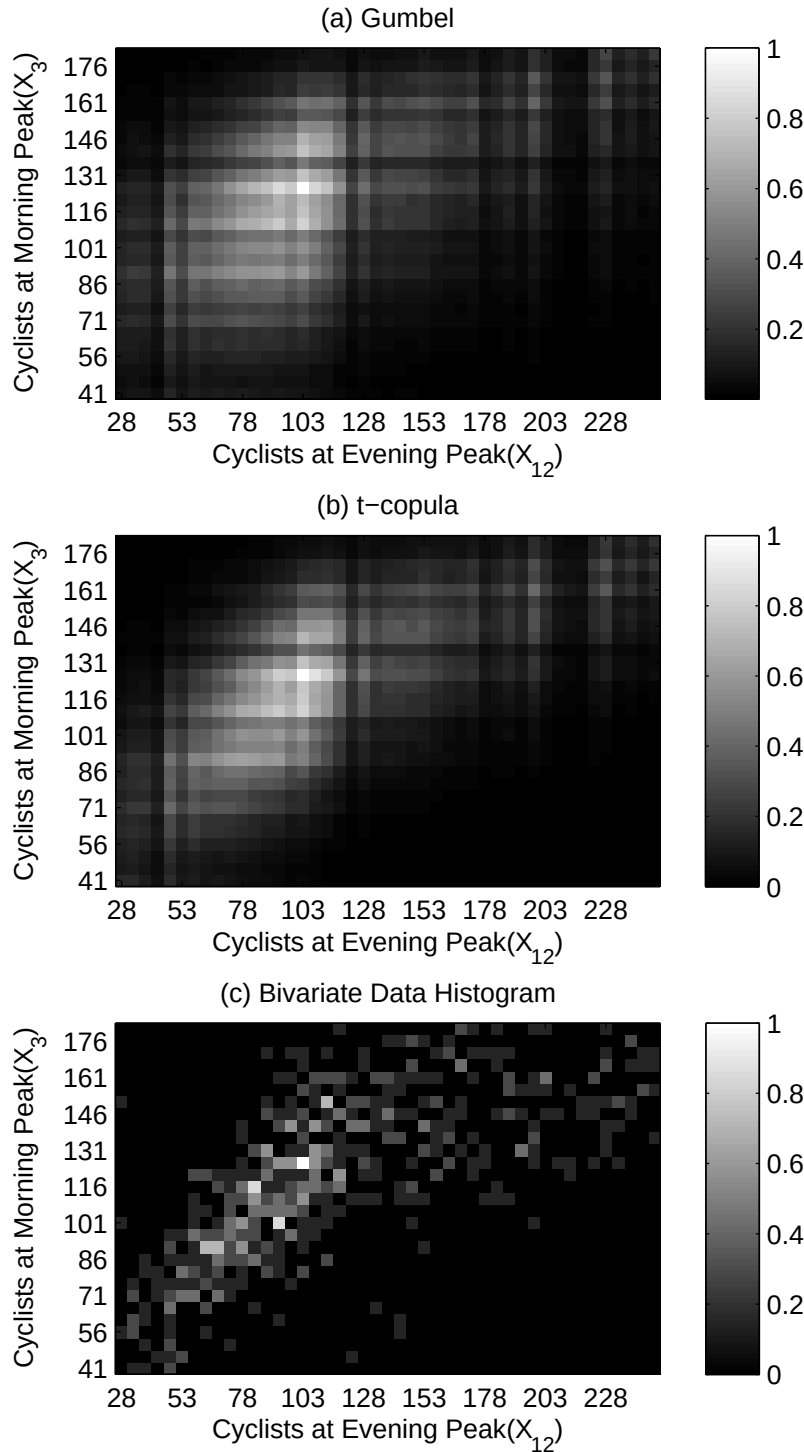


Figure 4: Panels (a) and (b) are the estimated bivariate marginal probability mass functions $f_{3,12}(X_3, X_{12})$ arising from the 16 dimensional D-vines with (a) Gumbel pair-copulas and (b) t pair-copulas. The mass functions are normalized to $[0, 1]$ and binned for presentation, with bin widths 3 and 5 for X_3 and X_{12} , respectively. The univariate margins $F_3(X_3)$ and $F_{12}(X_{12})$ are the empirical distribution functions, which produce the ‘stripey’ effects. Panel (c) is a bivariate (normalized) histogram of the observed counts X_3 and X_{12} for comparison.